

Markerless Mixed-Reality Injection Training on Commodity Headsets: Real-Time 3D Geometric Guidance for Needle Angle and Depth

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[PLACEHOLDER]

Teaser: in-headset view of the markerless injection trainer. A virtual syringe (reconstructed from hand tracking), the learner-placed injection site, the blue valid-angle cone for the selected injection type, and the green/red/orange depth zones below the skin. Replace with a composite passthrough capture.

Figure 1: The markerless MR injection trainer running on a standalone Meta Quest 3. No haptic device, instrumented prop, or external tracker is used.

Abstract

Safe injection technique—correct entry angle, insertion depth, and steadiness—is a high-volume clinical skill that benefits from repeated, low-risk practice. Existing virtual- and mixed-reality (MR) needle-insertion trainers achieve high fidelity with haptic devices, instrumented props, and external trackers, but this hardware raises cost and setup burden and limits deployment to a handful of equipped labs. We present a *markerless, hardware-free* MR injection trainer that runs entirely on a consumer standalone headset (Meta Quest 3). The system reconstructs a virtual syringe from articulated hand tracking and overlays *real-time 3D geometric guidance* on a learner-placed injection site: a colour-coded cone of valid entry angles specific to the chosen injection type (intramuscular, subcutaneous,

intradermal, intravenous) and stacked depth zones (correct, too-deep, maximum) that the needle must respect. To counter the angle and depth error introduced by finger occlusion of the held syringe, we introduce a *center-snap* pose model that anchors the needle entry to the target site and derives angle from a long, stable lever. The trainer scores each attempt on angle, depth, speed, flow, and removal stability. We describe the system architecture and a study protocol that evaluates accuracy (angle error, depth error, tip-to-center distance, completion time) and usability (SUS, NASA-TLX) with and without the guidance overlay. [PLACEHOLDER RESULTS: insert headline findings after the study is run.]

CCS Concepts

• **Human-centered computing** → Mixed / augmented reality; Empirical studies in HCI; • **Applied computing** → Health informatics; Interactive learning environments.

Keywords

mixed reality, hand tracking, medical simulation, injection training, needle insertion, visual guidance, standalone HMD, Meta Quest

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1 Introduction

Parenteral injection is among the most frequently performed clinical procedures, and poor technique—wrong entry angle, shallow or excessive depth, or an unsteady hand—contributes to pain, tissue damage, and failed delivery. Competence is built through repetition, but live practice is constrained by patient safety, consumables, and supervisor availability. Simulation addresses this gap, and virtual reality (VR) training has repeatedly transferred to improved real-world performance in adjacent procedural skills [5].

Most VR/MR needle trainers pursue *physical fidelity*: they couple a head-mounted display with haptic devices, instrumented syringes, or electromagnetically tracked tools so that learners feel tissue resistance and manipulate a real object [2, 4, 7]. This fidelity is valuable, but the hardware is expensive, requires per-session calibration across coordinate systems, and ties the experience to a small number of equipped labs. The result is a training modality that scales poorly to the large, distributed populations—nursing and medical students, community health workers—who most need volume practice.

We take the opposite design stance and ask how much useful injection-technique training can be delivered with *no added hardware at all*: a single consumer standalone headset, markerless articulated hand tracking, and software. Rather than reproduce haptic sensation, we make the *geometry of a correct injection explicit and continuously visible*. The learner pinches to place an injection site; the system reconstructs a virtual syringe from their hand and renders, in 3D and in real time, (i) a cone of acceptable entry angles whose aperture is specific to the selected injection type, and (ii) stacked depth zones—correct, too-deep, and a maximum-depth floor—that the needle tip must respect (Figure 1). Each attempt is scored on angle, insertion depth and speed, plunger flow rate, and withdrawal stability.

A core technical obstacle is that the hand holding a syringe *occludes the very fingers used to infer the tool's pose*, so a naïvely reconstructed needle axis and tip jitter badly. We contribute a *center-snap* pose model: because the learner is always aiming at the placed site, we anchor the needle's entry point to the site center and derive the syringe angle from the long, stable lever between that anchor and the hand, which is far less sensitive to fingertip noise than the short, occluded needle segment.

Contributions.

- A markerless, hardware-free MR injection-technique trainer that runs entirely on a commodity standalone HMD, lowering the cost and setup barrier of prior haptic/instrumented systems.
- A real-time 3D geometric guidance design—per-injection-type valid-angle cones and colour-coded depth zones—that externalizes the clinical targets for angle and depth.
- A center-snap pose model that makes hand-tracked angle and depth estimation robust to the finger occlusion inherent in holding a virtual syringe.
- A study protocol and [PLACEHOLDER: results] evaluating accuracy and usability, with and without guidance.

2 Related Work

2.1 VR and MR needle-insertion training

Needle and injection trainers have been a recurring target for immersive simulation. Recent systems emphasize haptic realism: Kim et al. [4] integrate two heterogeneous haptic devices with a HoloLens 2 for bimanual intravenous insertion, varying skin and vein properties, and evaluate with 20 participants (9 experts, 11 novices) using the NASA-TLX. Guruge et al. [2] combine MR with haptic feedback for acupuncture needling. These works deliver convincing tactile cues but depend on specialized devices and multi-coordinate calibration. Our system deliberately forgoes haptics to remain deployable on unmodified consumer hardware, trading tactile fidelity for accessibility and continuous *visual* guidance.

2.2 Hand tracking and tool guidance

Markerless articulated hand tracking is now standard on consumer headsets, enabling controller-free interaction but introducing jitter and occlusion when the hand grips a tool. Mixed-reality procedural simulators have used external cameras and spatial co-registration to track hands together with instrumented tools [6], and multi-modal approaches add wrist-based vibrotactile cues to AR overlays for handheld-tool guidance [7]. We instead reconstruct the tool purely from on-device hand joints and address occlusion in software via the center-snap model.

2.3 Guidance cues and scoring metrics

Needle-training systems commonly report composite accuracy from normalized metrics—insertion depth, angular deviation, tip-to-center distance, and task duration—and contrast novice and expert performance. Depth-responsive visual guidance has been shown to reduce mean depth error and task time in MR needle tasks. Our guidance makes these same target quantities (angle window, depth band, entry point) *directly visible as 3D geometry* rather than as numeric read-outs, and our scoring adopts the established angle/depth/time metrics so results are comparable to prior work.

2.4 Positioning

Table 1 contrasts representative systems on hardware, headset class, feedback modality, and accessibility. To our knowledge, ours is the first injection-technique trainer that combines markerless hand-tracked tool reconstruction, per-type 3D geometric guidance, and on-device scoring on a single commodity standalone HMD with no added hardware.

3 System

3.1 Architecture

The trainer is built in Unity and runs as a standalone Android application on Meta Quest 3 via OpenXR and the Unity XR Hands subsystem; mixed-reality passthrough places the virtual content in the user's real environment. The runtime is organized as a small set of cooperating components (Figure 2): SyringeOverlayTracker reconstructs the syringe from hand joints; SurfaceSelectionTool captures the injection site from pinch gestures; InjectionTargetGuide renders the 3D guidance; SyringeCalibrationButtonBridge holds

Table 1: Representative needle/injection trainers vs. our system. [PLACEHOLDER: verify each cell against the cited source before camera-ready.]

System	Headset / display	Tool tracking	Feedback	Added hardware
Kim et al. [4]	HoloLens 2 (MR)	Haptic gloves + stylus	Haptic + visual	Yes (high)
Guruge et al. [2]	MR display	Instrumented	Haptic + visual	Yes (high)
Yang et al. [7]	AR	Tracked handheld tool	Wrist haptic + AR	Yes (moderate)
SpatialTutor [6]	MR + external cameras	CV + co-registration	Visual	Yes (moderate)
Ours	Quest 3 (standalone)	Markerless hand tracking	3D visual guidance	None

the per-type clinical profiles and computes the per-attempt metrics and score; and GoalManager orchestrates the guided workflow and the floating coaching UI. No physical syringe, marker, or external sensor is used.

3.2 Hand-tracked syringe and occlusion robustness

The virtual syringe is reconstructed each frame from thumb, index, and middle finger joints: the grip defines the barrel axis, from which the wings, barrel tip, and needle tip are extrapolated using fixed syringe dimensions (barrel 60 mm; needle 47 mm beyond the barrel tip). Two problems arise from markerless tracking. First, when the thumb or fingers are momentarily occluded the reconstructed axis can *flip*; we stabilize its sign against a non-invertible wrist→middle-finger reference so the needle always points out of the hand. Second, and more importantly for scoring, the short needle segment is exactly where occlusion noise is worst.

Center-snap pose. Because the task is always “inject at the placed site,” we anchor the needle entry point to the site center once the tip first reaches it, and measure *insertion depth* as travel along the ideal injection axis from that anchor, and *entry angle* from the long lever between the anchor and the hand (plunger) rather than from the noisy needle tip. This trades a small, deliberate modeling assumption (the needle enters at the target center) for substantially steadier angle and depth estimates under occlusion. Section 4 evaluates the accuracy this yields.

3.3 Real-time 3D geometric guidance

After the learner places a site (three pinch points define a small surface and its normal), the InjectionTargetGuide renders three coupled cues anchored at the site (Figure 3):

- **Injection spot** — a disc at the site center that shrinks (10 mm → 4 mm) as the needle approaches, focusing aim.
- **Valid-angle cone (blue/green)** — a slant band swept around the surface normal whose inner and outer edges are the steepest and shallowest acceptable entry angles for the selected injection type. The band turns green while the live syringe angle lies inside the valid range.
- **Depth zones** — a tapering channel along the ideal needle path: a *green* segment up to the correct depth, a *red* segment for over-insertion, and an *orange* disc marking the maximum safe depth.

The cone is sized by a fixed slant length so its on-skin footprint stays readable across injection types rather than ballooning for

Table 2: Configured per-type profiles in the trainer (entry angle from the skin surface; depths along the needle path). [PLACEHOLDER: confirm clinical values.]

Type	Angle range	Correct depth	Max depth
Intradermal (ID)	8–18°	2 mm	4 mm
Subcutaneous (SC)	35–50°	8 mm	12 mm
Intramuscular (IM)	80–95°	25 mm	38 mm
Intravenous (IV)	20–35°	10 mm	15 mm

shallow angles, and it can be previewed in front of the user while selecting injection type, before a site is placed.

3.4 Per-type profiles and scoring

Each injection type carries a profile of an acceptable entry-angle range and a correct/maximum needle depth (Table 2); these drive both the guidance geometry and the score. The configured defaults are starting points exposed for tuning and **should be validated against clinical guidelines and a clinician co-author before camera-ready**. For every attempt the system computes angle relative to the surface, insertion depth along the ideal axis, axial insertion speed, lateral (steadiness) speed, plunger flow rate, and withdrawal speed/stability, and combines them into a per-category ScoreBreakdown.

3.5 Workflow and coaching UI

A floating, grabbable coaching panel guides the learner through the procedure—select injection type, place the site, achieve and hold a valid angle, insert to the correct depth at a steady speed, dispense, and withdraw—advancing step by step and surfacing live metrics and a final score. The panel can be world-pinned or follow the head, and the syringe and hand-skeleton overlays can be toggled. [PLACEHOLDER: coaching-UI screenshot.]

4 Study

[This section is a pre-registered-style protocol; results are PLACEHOLDERS to be filled after the study runs (planned before the camera-ready deadline).]

4.1 Design and participants

We will recruit [N = TODO, target ≈16] participants spanning novices (students with no injection experience) and a smaller set

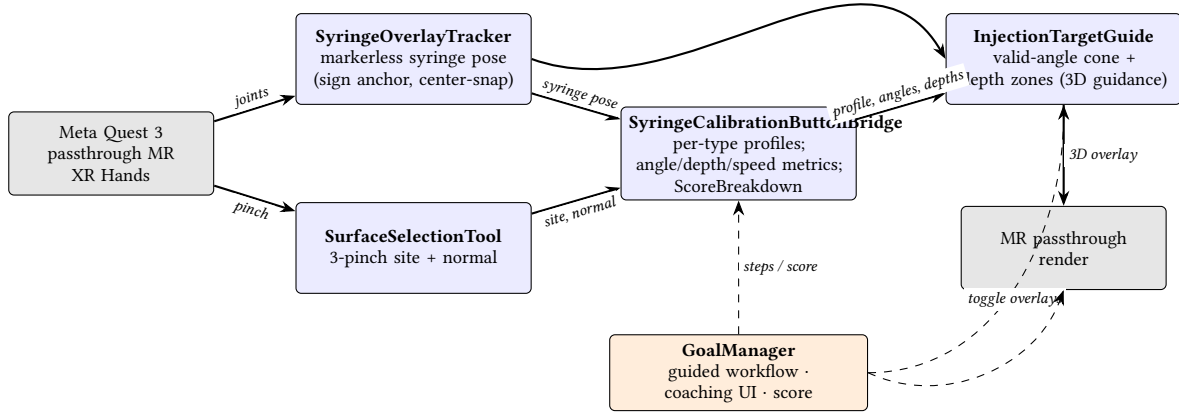


Figure 2: System architecture and data flow. Hand-tracking feeds markerless syringe reconstruction (**SyringeOverlayTracker**) and pinch-based site placement (**SurfaceSelectionTool**); the **SyringeCalibrationButtonBridge** holds per-type clinical profiles and derives the geometric metrics and score; **InjectionTargetGuide** renders the 3D guidance; and **GoalManager** orchestrates the workflow and coaching UI (dashed = control/visibility). No haptic device, marker, or external tracker is used.

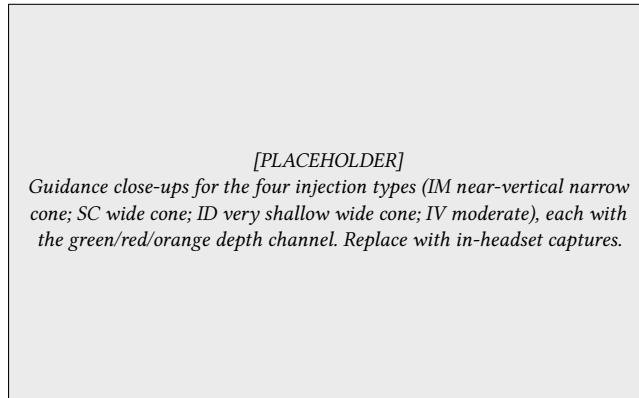


Figure 3: Per-injection-type 3D guidance: valid-angle cone (turns green when the held syringe is in range) and colour-coded depth zones.

of clinically experienced users. The primary comparison is within-subject *guidance ON vs. OFF*; injection type is counterbalanced. The study is conducted under [IRB/ethics approval no. TODO].

4.2 Apparatus and task

Meta Quest 3, standalone, with passthrough; no other hardware. After a short familiarization, each participant performs a fixed set of injection attempts per type, with and without the guidance overlay, on a learner-placed virtual site. [PLACEHOLDER: apparatus photo.]

4.3 Measures

Accuracy (logged by the system): entry-angle error (°) from the type’s target range, insertion-depth error (mm) from the correct

Table 3: Per-type accuracy and usability summary. [PLACEHOLDER: fill with real data.]

Measure	ID	SC	IM	IV
Angle error (°)	–	–	–	–
Depth error (mm)	–	–	–	–
Tip-to-center (mm)	–	–	–	–
Completion time (s)	–	–	–	–

depth, tip-to-center distance (mm), and completion time (s). **Usability:** System Usability Scale (SUS) [1] and NASA-TLX workload [3], plus open-ended feedback. These accuracy metrics match those reported in prior needle-training work for comparability.

4.4 Procedure and analysis

Familiarization, then counterbalanced ON/OFF blocks across injection types, then questionnaires. We will report descriptive statistics and within-subject tests (repeated-measures ANOVA or Wilcoxon signed-rank, as appropriate) for the guidance effect on each accuracy measure and on workload, with effect sizes.

4.5 Results and findings

[PLACEHOLDER] Figure 4 and Table 3 report accuracy and usability outcomes. Headline findings (angle/depth error reduction with guidance, SUS score, NASA-TLX workload) go here once collected.

5 Discussion, Limitations, and Future Work

The system’s central trade-off is the absence of haptics: it cannot reproduce tissue resistance or the “pop” of a vein, so it targets *technique and procedural* learning (angle, depth, speed, sequence)

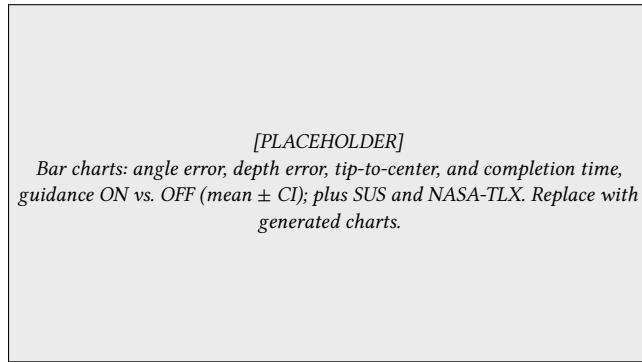


Figure 4: Accuracy and usability results (guidance ON vs. OFF). [PLACEHOLDER.]

rather than tactile discrimination, and is best seen as complementary to haptic stations rather than a replacement. The center-snap model assumes entry at the site center, which is appropriate for guided practice but would need relaxing for free-form placement assessment. Markerless tracking still degrades under severe occlusion and outside the headset’s tracking volume. The per-type profiles require clinical validation. Finally, our planned study measures in-simulation accuracy and usability, not transfer to real injections; a transfer study with clinician assessment is important future work, alongside skin-deformation cues and audio feedback.

6 Conclusion

We presented a markerless, hardware-free mixed-reality injection trainer for commodity standalone headsets that turns the clinical targets of a correct injection—entry angle and depth—into continuously visible 3D guidance, and that stabilizes hand-tracked pose against occlusion with a center-snap model. By removing haptic and instrumented hardware, the system trades tactile fidelity for accessibility and scale. **[PLACEHOLDER: one-sentence headline result.]**

Ethics Review

[PLACEHOLDER] The user study was reviewed and approved by **[Institutional Ethics Committee, approval no. TODO]**. Participation is voluntary and informed; no patient data are used; the system is a training aid and not a medical device.

Acknowledgments

[PLACEHOLDER] We thank the participants and **[funding/lab acknowledgments TODO]**.

Code and Demo Video

Source code: **[repository URL TODO]**. Demo video: **[unlisted video URL TODO]**.

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